



Bug Bounty Report

**LimitOrderLib: Zero-Collateral Partial Fills Fully
Consume Signed Sell Orders Without Payment**

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Summary

`LimitOrderLib.validateLimitOrderFairness()` permits partial fills in which synthetic assets change hands while the collateral payment rounds down to zero. A signed sell order for 100 synthetic against 3 collateral can be filled in four transactions ($33 + 33 + 33 + 1$), transferring all 100 synthetic to the buyer for no payment and fully consuming the seller's signed order.

Severity: Critical

Affected code: `src/orderbook/LimitOrderLib.sol`, lines 81-107 and 180-181

Root Cause

The buy-side fairness check explicitly accepts a zero sold amount:

```
1  if (self.isBuyingSynthetic) {
2      int256 actualSold = actualCollateral;
3      // ...
4      if (actualSold == 0) {
5          return;
6      }
7  }
```

The sell-side rounding formula also allows non-zero synthetic to be sold for zero collateral:

```
1  actualSold * amountBuy < amountSell * (actualBought + 1)
```

For an order selling 100 synthetic for 3 collateral, a fill of 33 synthetic for 0 collateral passes because:

```
1  33 * 3 < 100 * (0 + 1)
2  99 < 100
```

`updateFulfillment()` only limits cumulative synthetic fulfillment. It never requires the cumulative collateral received to satisfy the signed price, so the rounding loss can be repeated until the order is exhausted.

Happy Path vs. Attack Path

Happy Path

1. A seller signs an order to exchange 100 synthetic for 3 collateral.
2. Matching fills transfer synthetic from the seller and collateral from the buyer.
3. Cumulative fulfillment reaches 100 synthetic only after the seller receives the signed consideration.

Attack Path

1. The victim signs a sell order for 100 synthetic against 3 collateral.
2. The attacker controls a matching buy order.
3. The authorized batcher submits four fills with `actualCollateral` = 0 and `actualSynthetic` = 33, 33, 33, 1.
4. Every fill passes both parties' fairness checks.
5. The seller order is fully consumed.
6. The buyer receives 100 synthetic and pays 0 collateral.
7. The seller receives 0 collateral and is left with a negative 100 synthetic balance.

Impact

- The signed limit price is bypassed.
- The buyer acquires the full synthetic position without payment.
- The victim's order is irrevocably consumed without consideration.
- In spot/predict deployments using `NullMarginCheck`, the seller can be pushed into an uncollateralized short position and free synthetic can later be settled for collateral.

Proof of Concept

The Foundry test is located at:

```
1 findings/H-02-ZeroCollateralTrade/test.t.sol
```

The test `test_zeroCollateralPartialFills_exhaustSellOrderForFree` signs the 100/3 order, executes the four zero-collateral fills, and verifies:

```
1 buyer synthetic balance: 100
2 seller synthetic balance: -100
3 buyer collateral paid: 0
4 seller collateral gained: 0
5 seller fulfillment: 100
```

Fix Direction

Reject zero-collateral fills whenever synthetic changes hands:

```
1 function validateLimitOrderFairness(...) public pure {
2 +   require(actualCollateral > 0 || actualSynthetic == 0, "
   LimitOrderLib: zero collateral fill");
3   ...
4 }
```

The protocol should also enforce cumulative price fairness across all partial fills so repeated rounding cannot reduce total seller proceeds below the signed order price.

Report Boundary

The direct loss path depends on an authorized batcher submitting the crafted fills. The critical impact applies to deployments where the batcher path is reachable and the orderbook operates without a margin plugin; stronger trust or margin assumptions reduce likelihood but do not repair the signed-price invariant.